$$U_R = 0. \tag{19}$$

, (16)

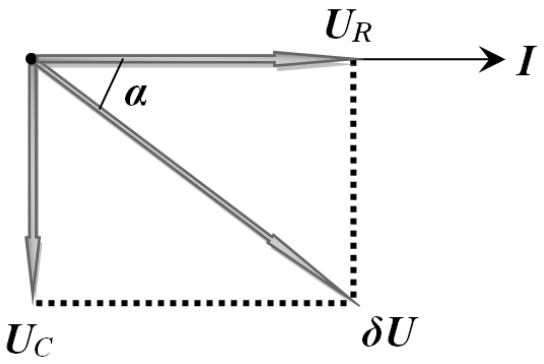
$$I = \frac{dq}{dt} = C(U)\frac{dU}{dt} + U\frac{dC(U)}{dU}\frac{dU}{dt} = C_{\text{eff}}\frac{dU}{dt}, \tag{20}$$

$$C_{\text{eff}} = C(U) + U\frac{dC(U)}{dU} = 0.200\ \mu F. \tag{21}$$

,

$$X_{\text{C}} = \frac{1}{\omega C_{\text{eff}}}. \tag{22}$$

.



$$I = \frac{\delta U}{\sqrt{R^2 + \frac{1}{\omega^2 C_{\text{eff}}^2}}} = 44.7\ \mu A. \tag{23}$$

α

$$\alpha = \text{arctg}\left(\frac{U_C}{U_R}\right) = \text{arctg}\left(\frac{1}{\omega C_{\text{eff}} R}\right) = 1{,}11\ rad = 63{,}4°. \tag{24}$$

$$I(t) = [44{,}7\sin(2500t + 1{,}1)]\ \mu A. \tag{25}$$

8. **[0.5점]**

$$U_C = \delta U \sin\alpha. \tag{26}$$

$$U_{\text{C}}(t) = U_{\text{C}} + \delta U \sin\alpha \sin\left(\omega t - \frac{\pi}{2} + \alpha\right) = \\ = [5{,}000 + 0{,}089\sin(2500t - 0{,}464)]\ V. \tag{27}$$

채점 기준

№	평가 항목	배점	
1	(1) $I = 0$	0.25	**0,75**
	(2) $C = 0{,}10\ \mu F$	0.25	
	(3) $q = CU_0 = 0.50\ \mu Cl$	0.25	
2	(4) $t = \infty$.	0.25	**0,25**
3	(5) $U_0 = \frac{q}{C} + IR$	0.25	
	(6) $I = \frac{dq}{dt}$	0.25	
	(7) $dt = \frac{R}{U_0 - \frac{q}{C(q)}}dq = f(q)dq$ $f(q) =$	0.25	